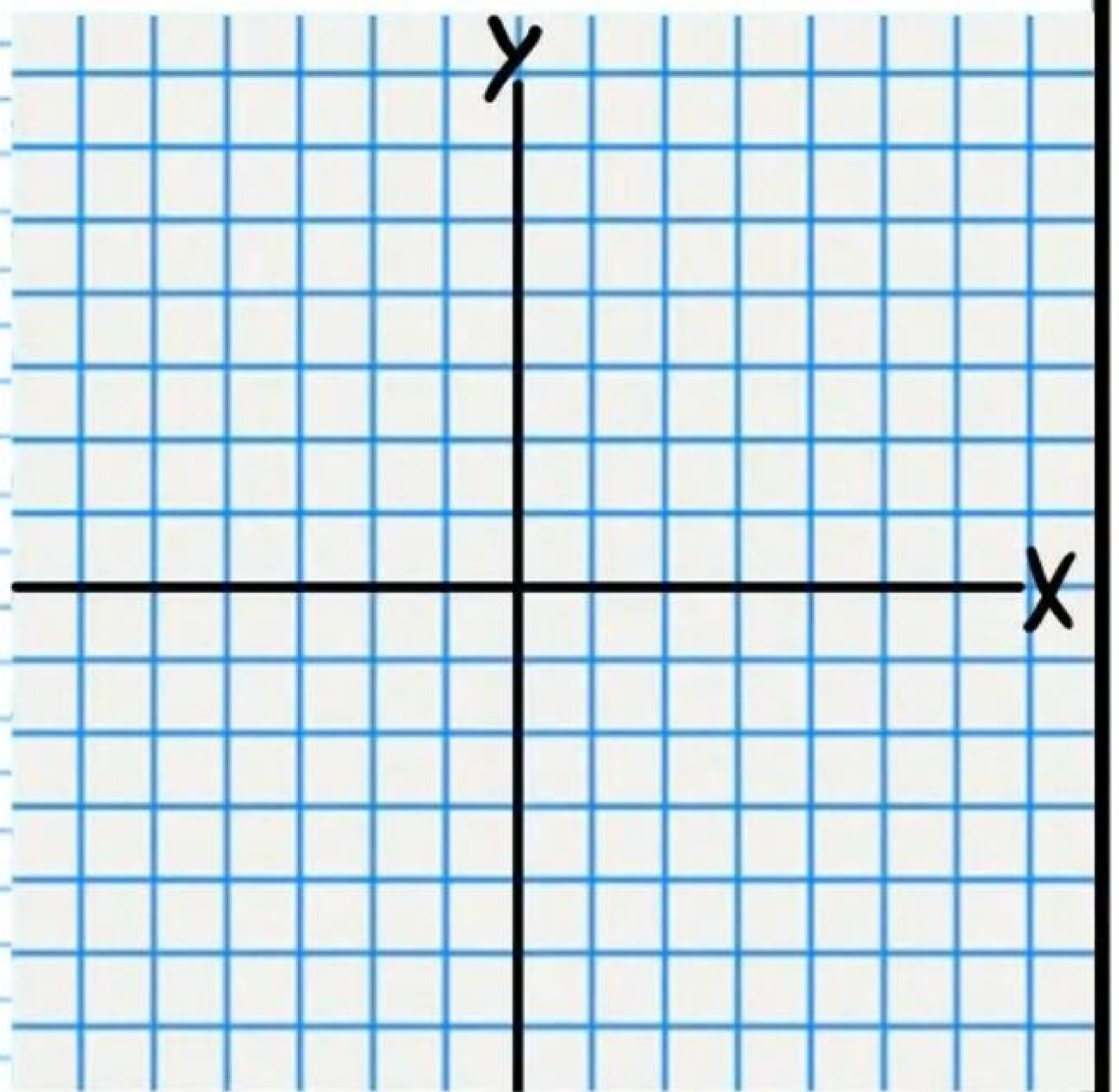
Name
Date
Course

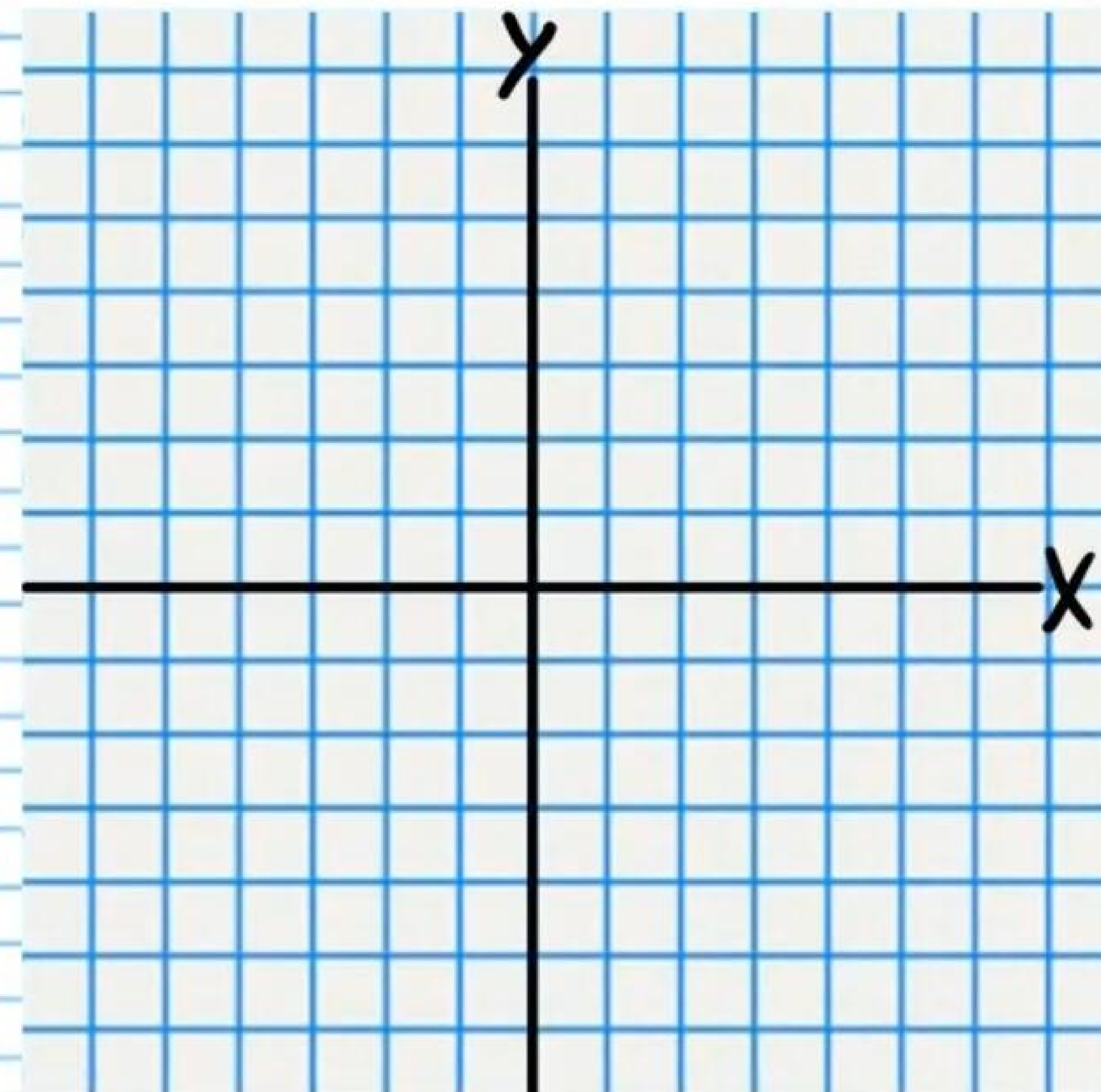
The Conics (Part III) Homework

Graph the following equations. ✱ You must first convert to a recognizable form.

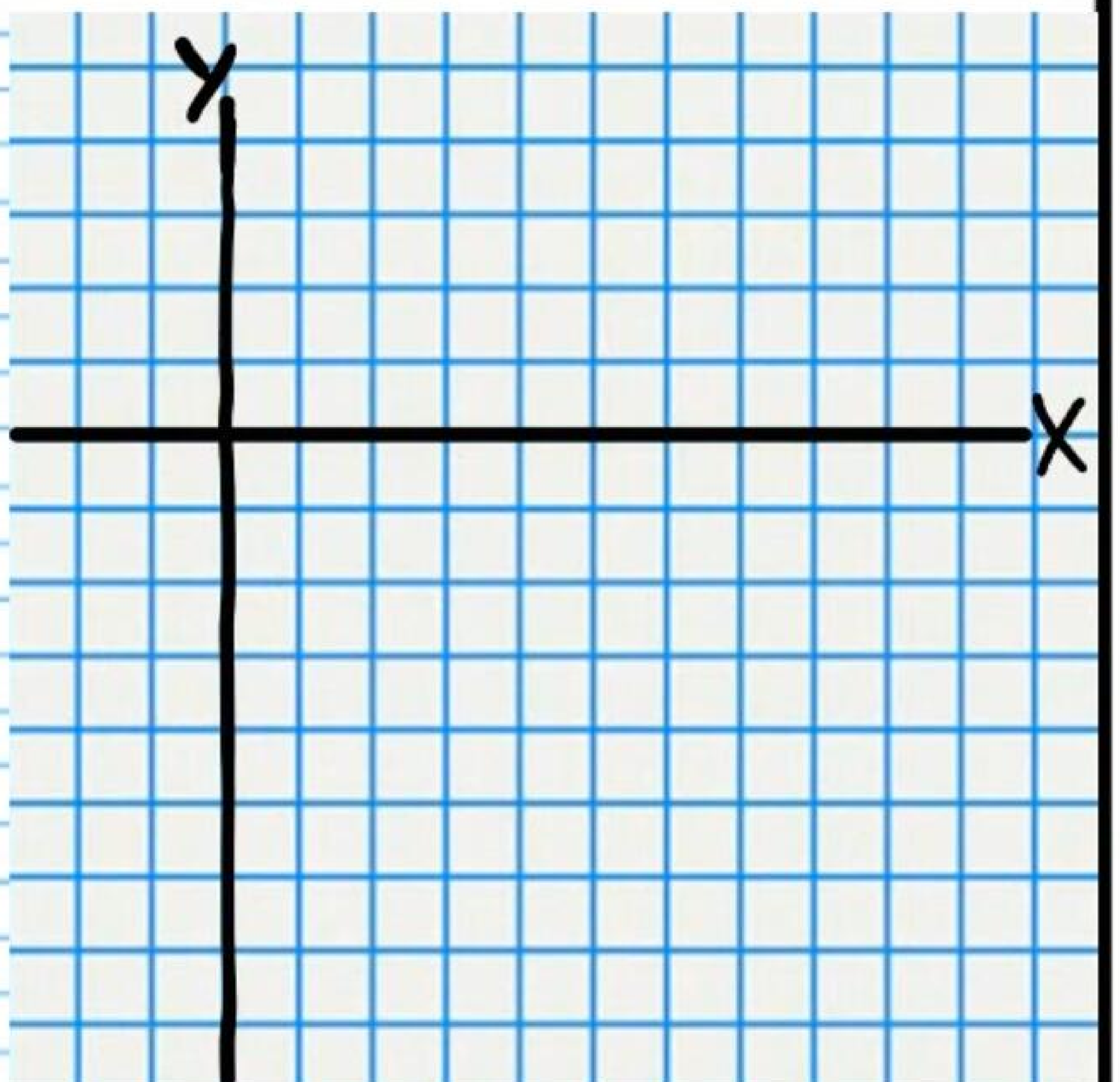
① $x^2+y^2+10x+24=0$



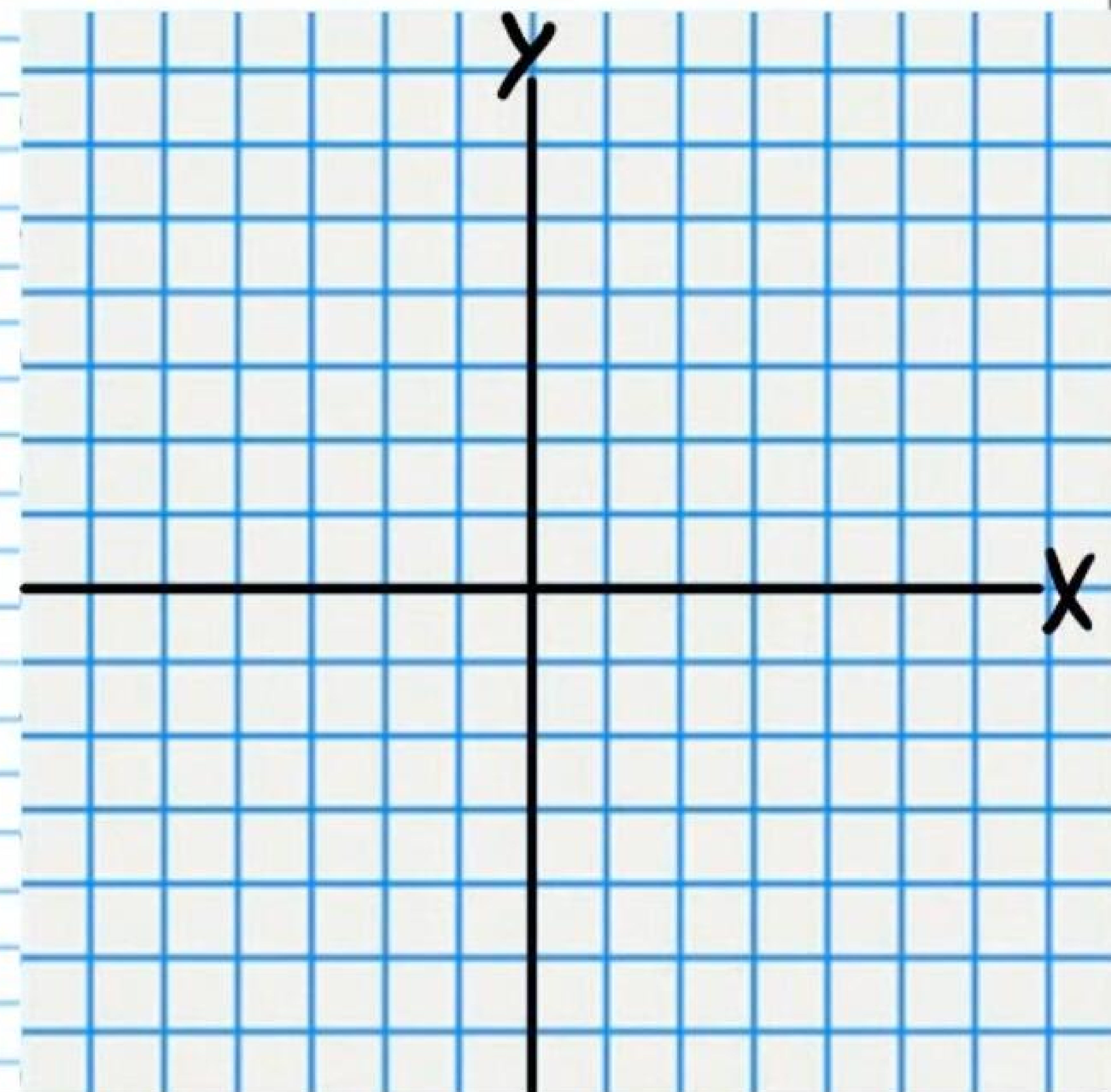
② $16x^2+9y^2-96x-18y+9=0$



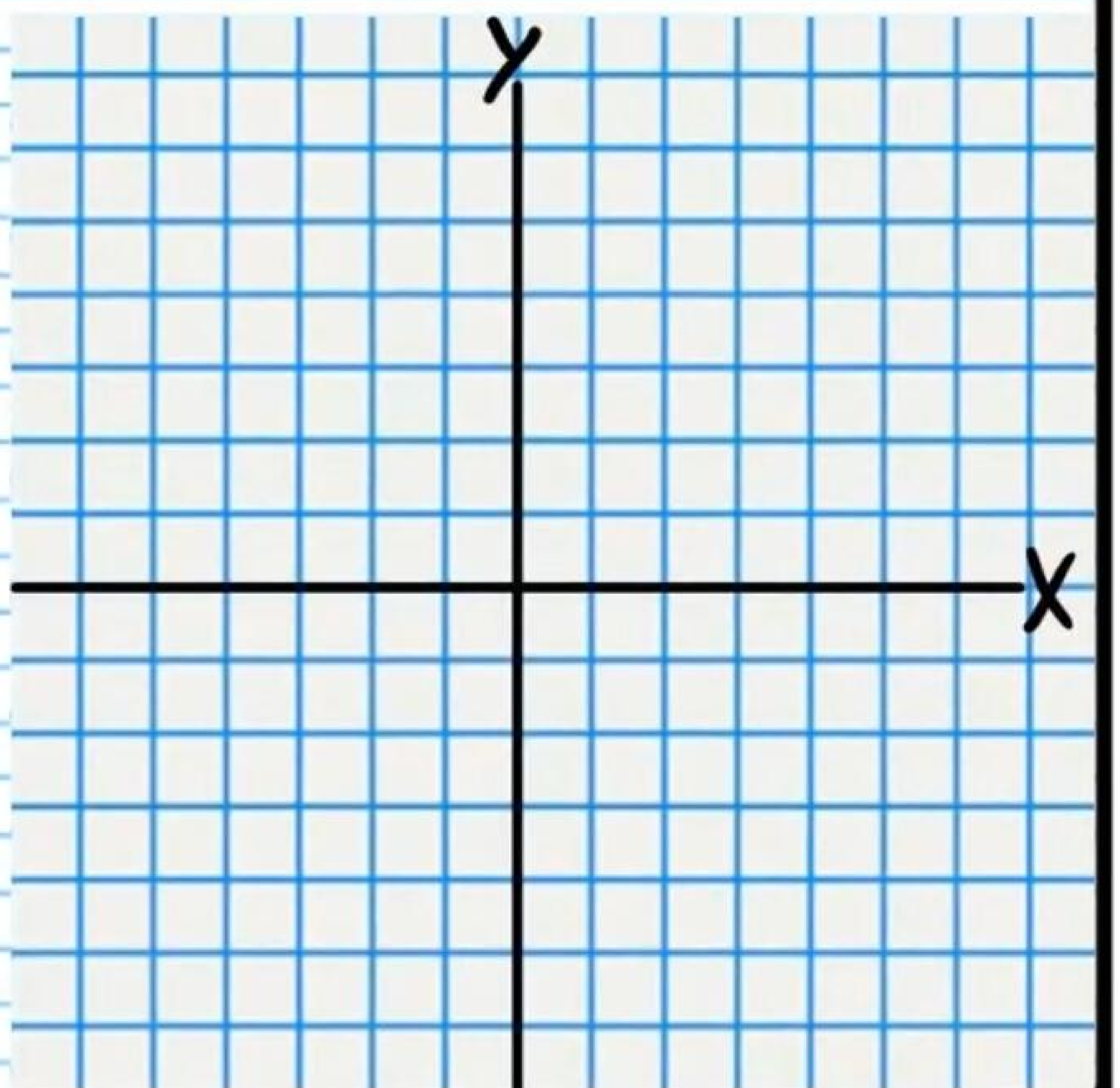
③ $y^2-x^2+4y+8x-37=0$



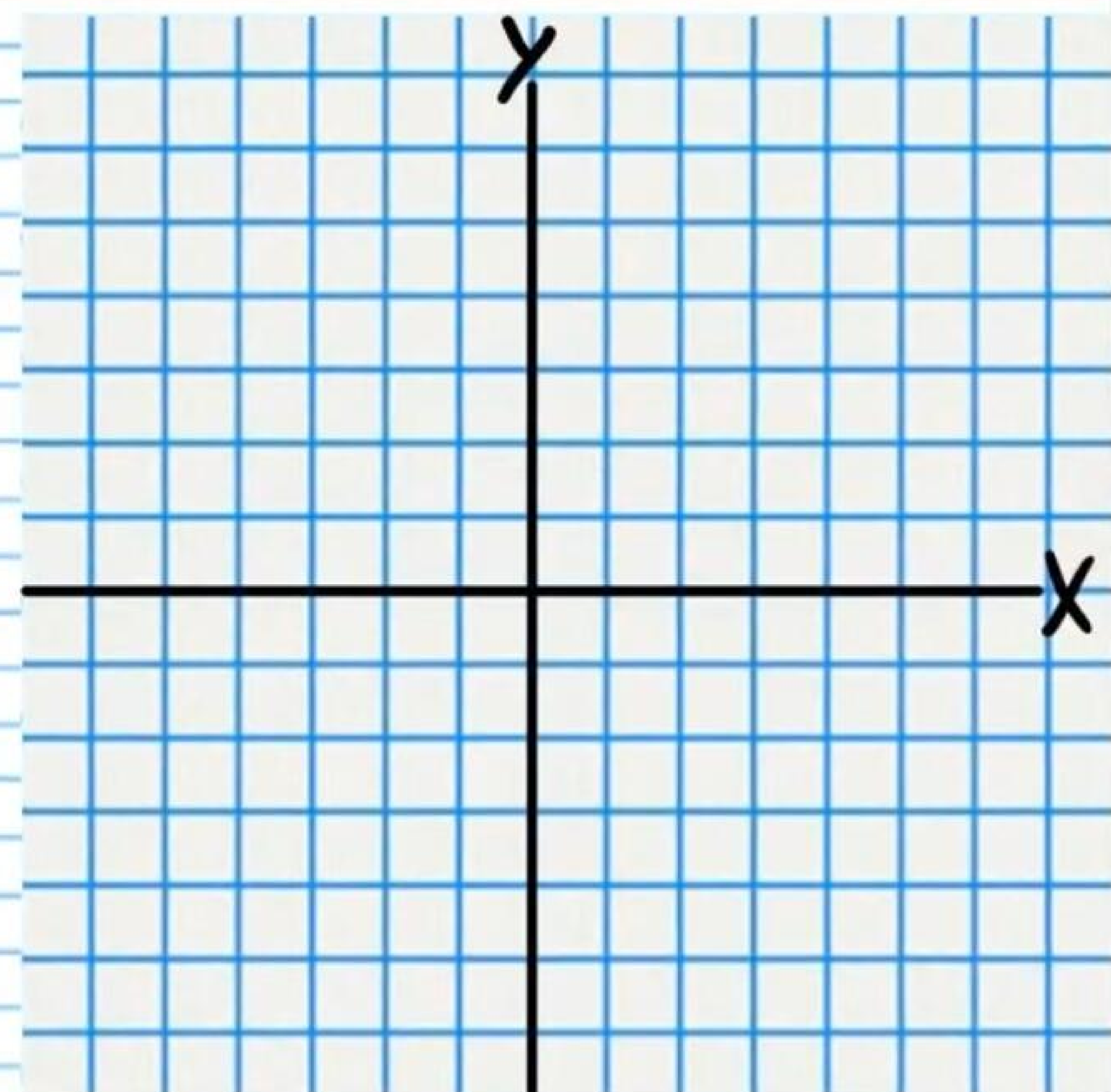
④ $3+4y=x^2+2x$



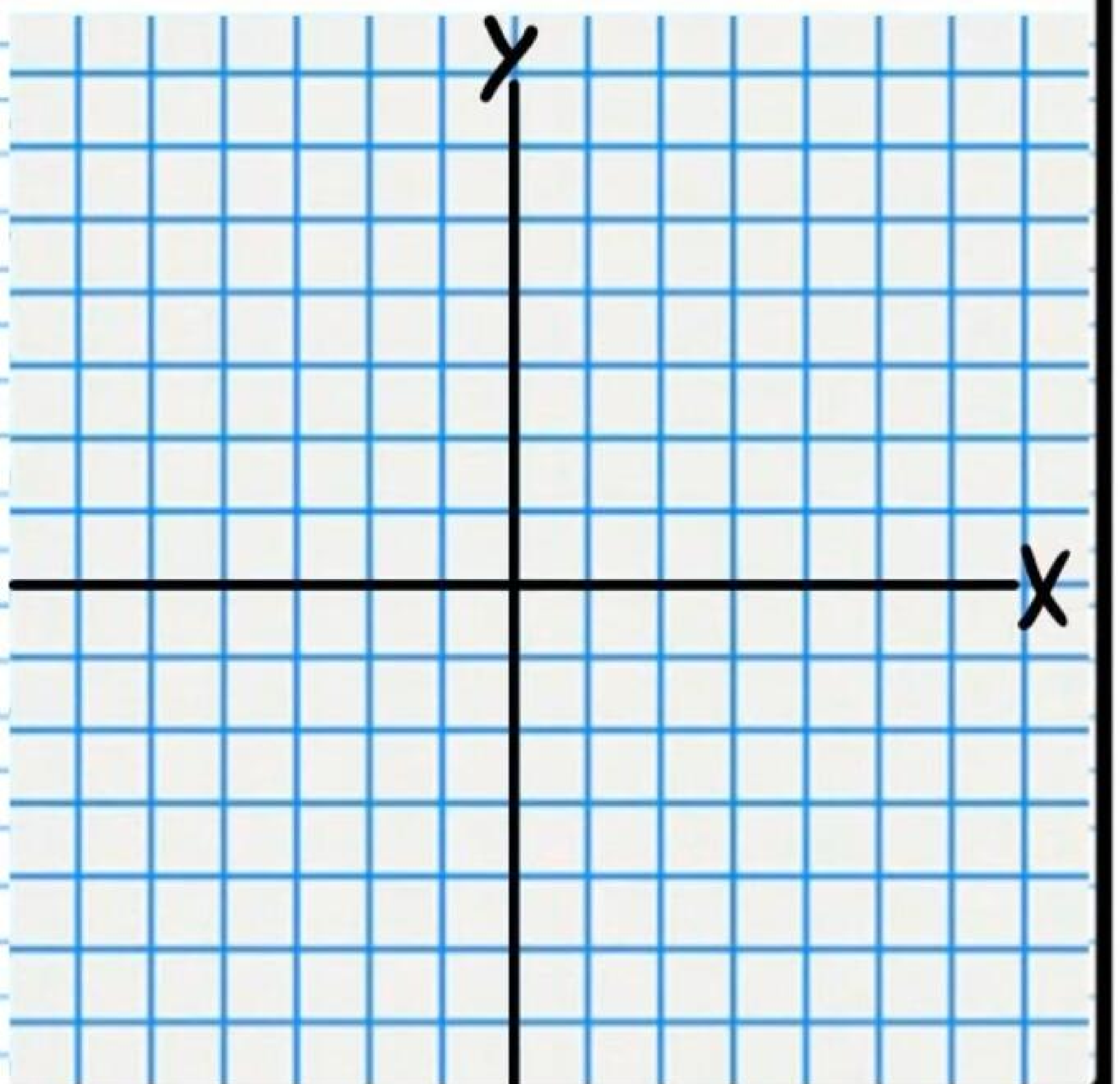
(5) $\frac{x^2}{4}+\frac{y^2}{4}=1$



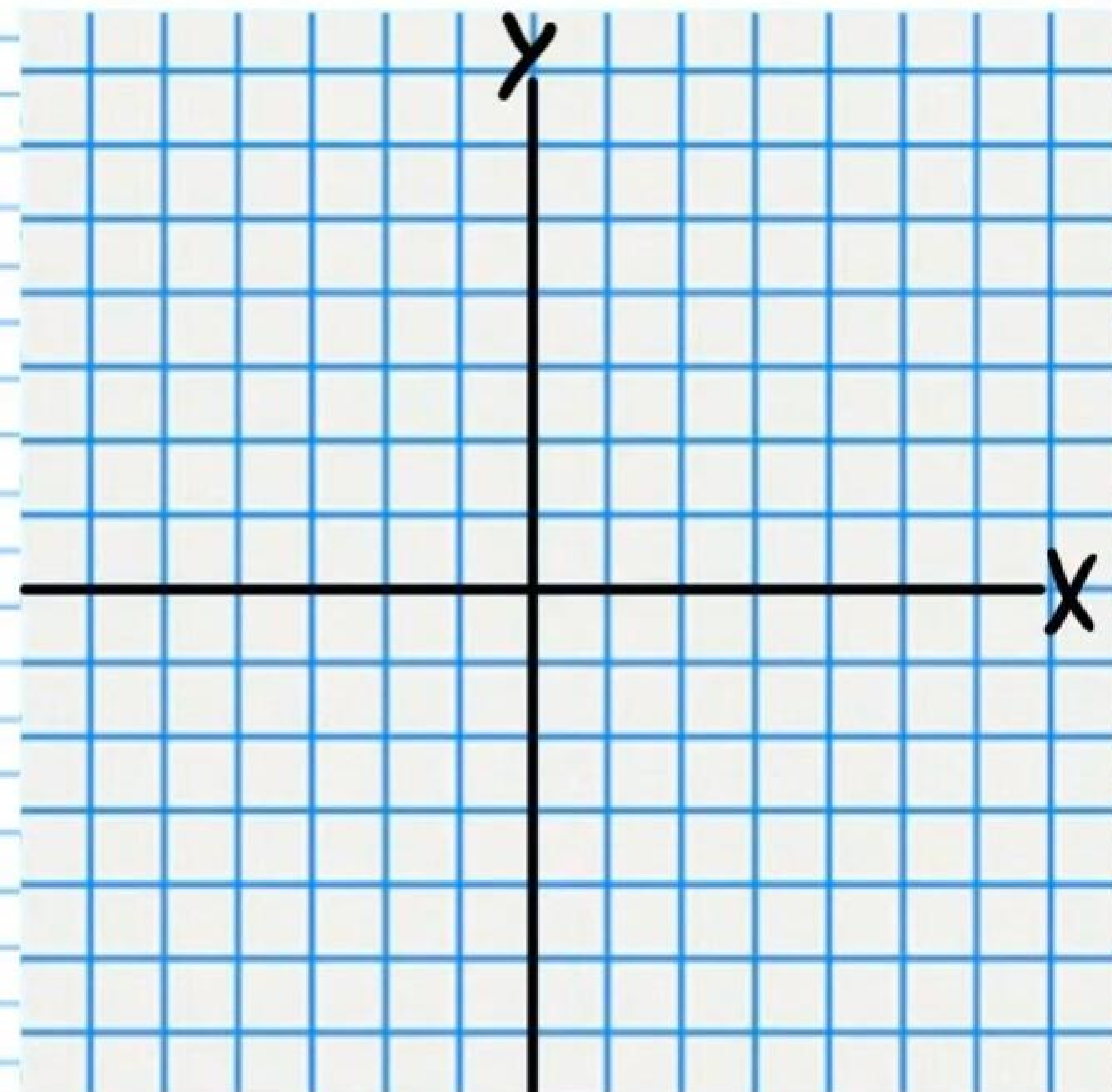
(6) $36x^2+y^2=36$



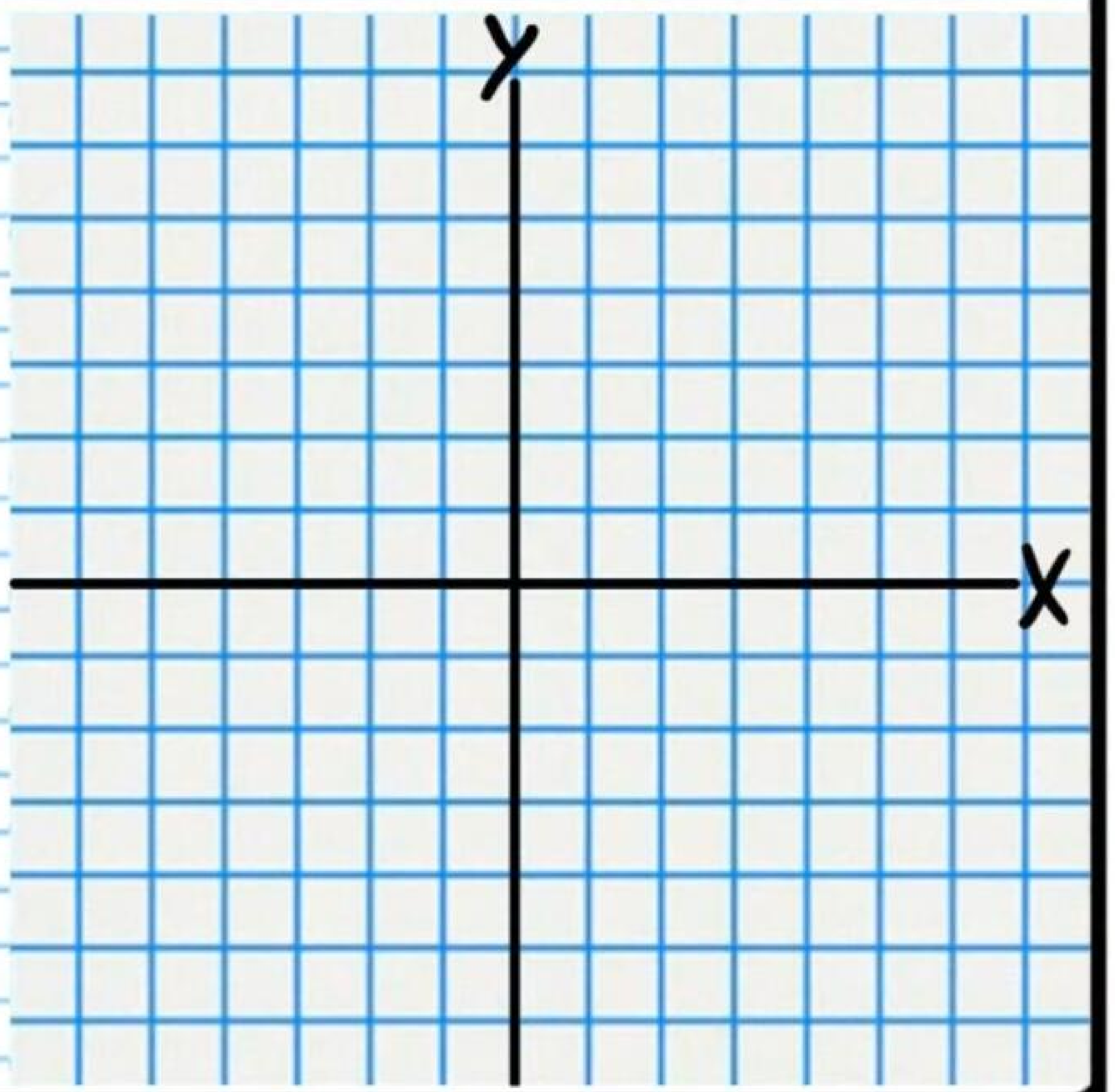
(7) $x = y^2 - 6y + 5$



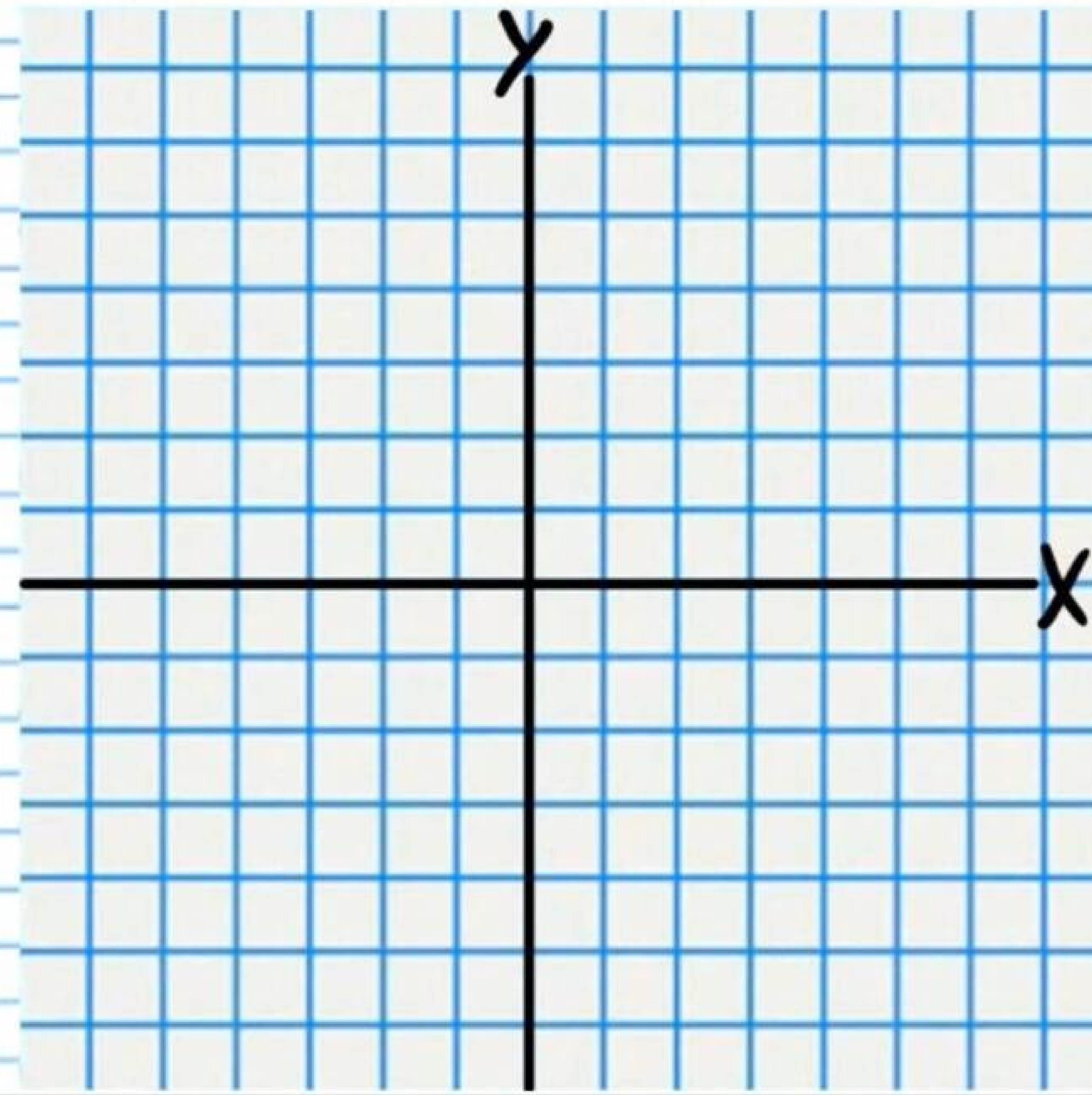
(8) $4x^2 + 36y^2 - 72y - 108 = 0$



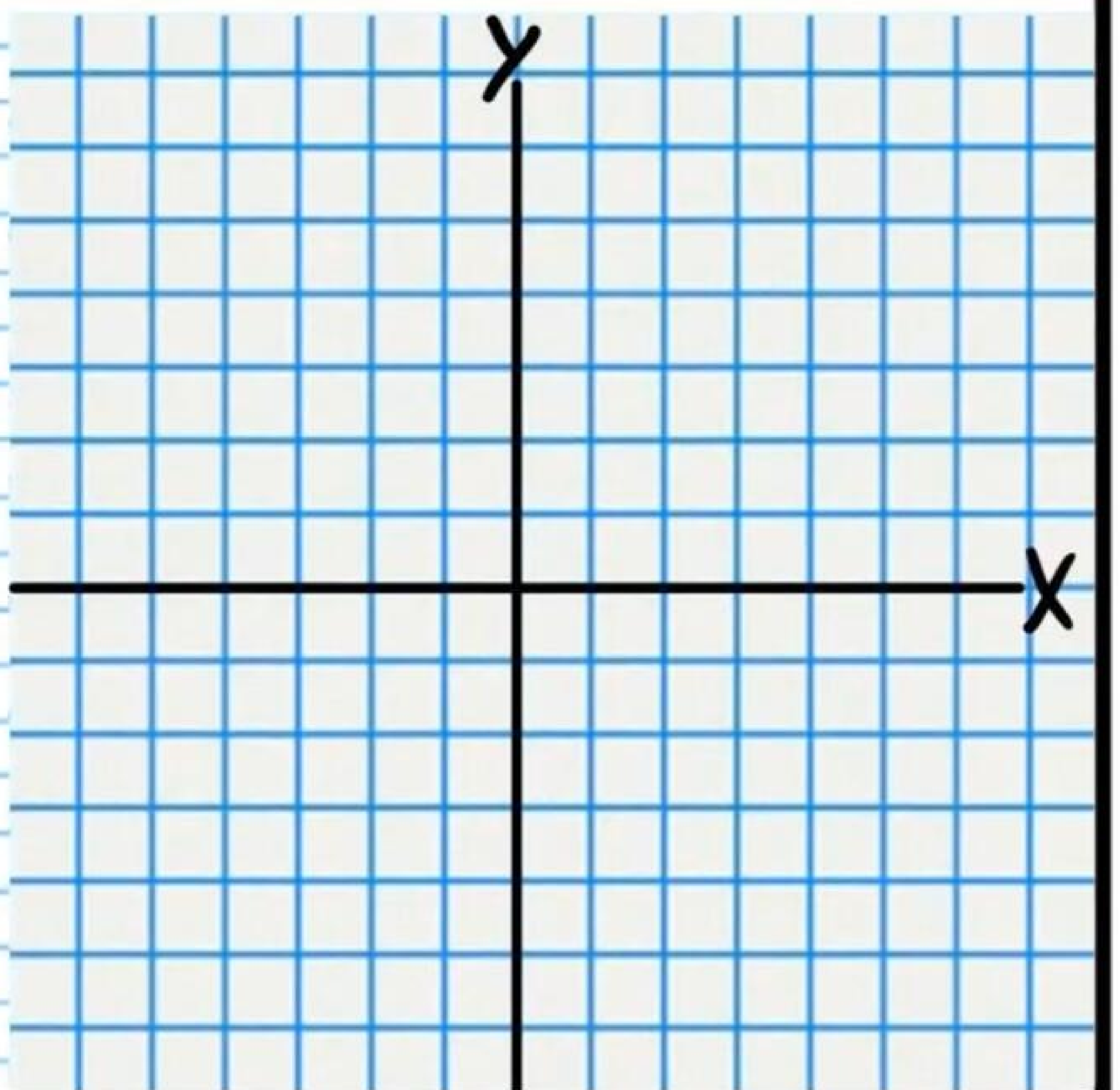
9 $-3x = y^2$



10 $y = -\frac{1}{2}x^2$

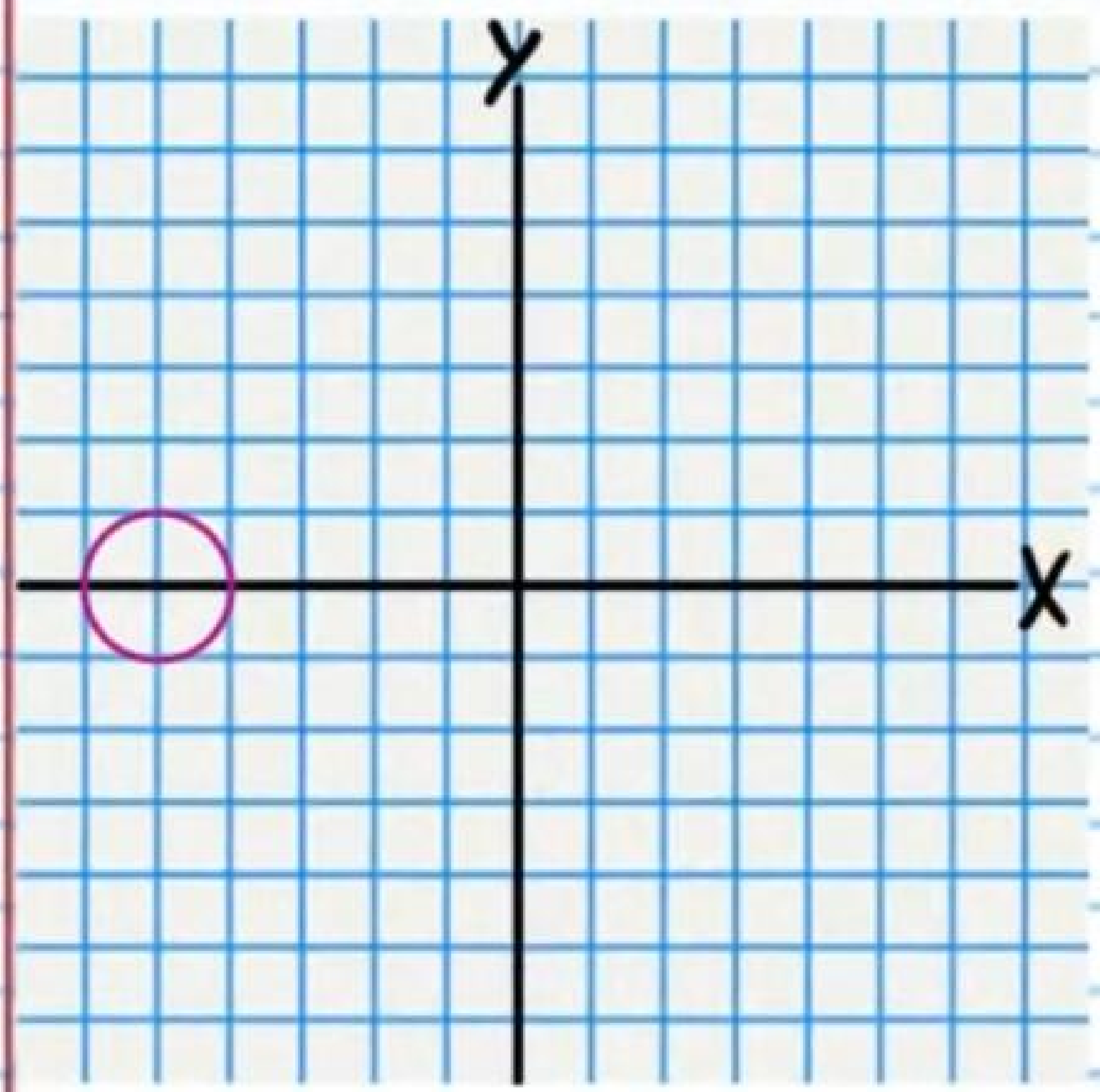


(11) $x^2-16y^2+128y-272=0$

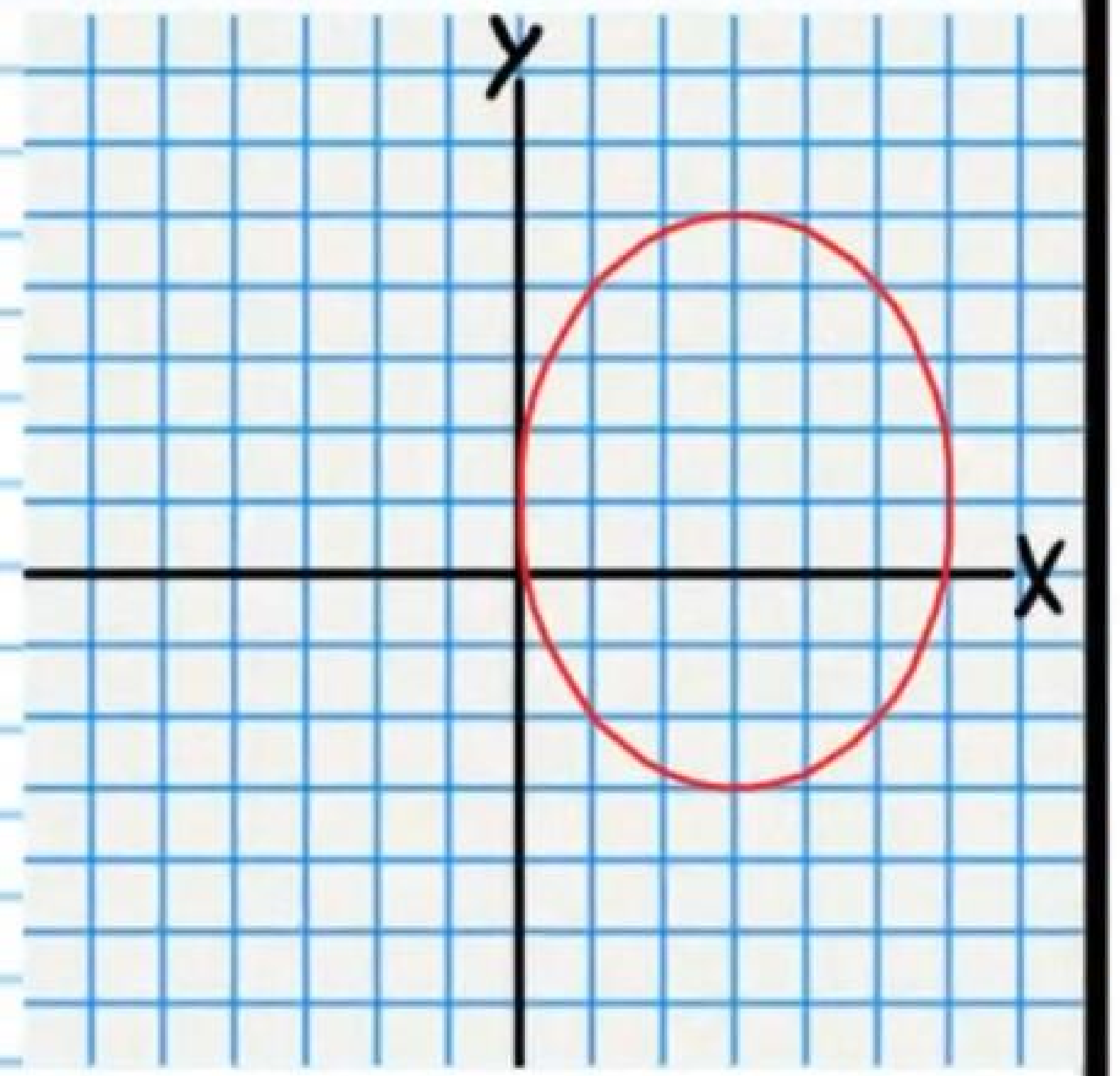


Answers

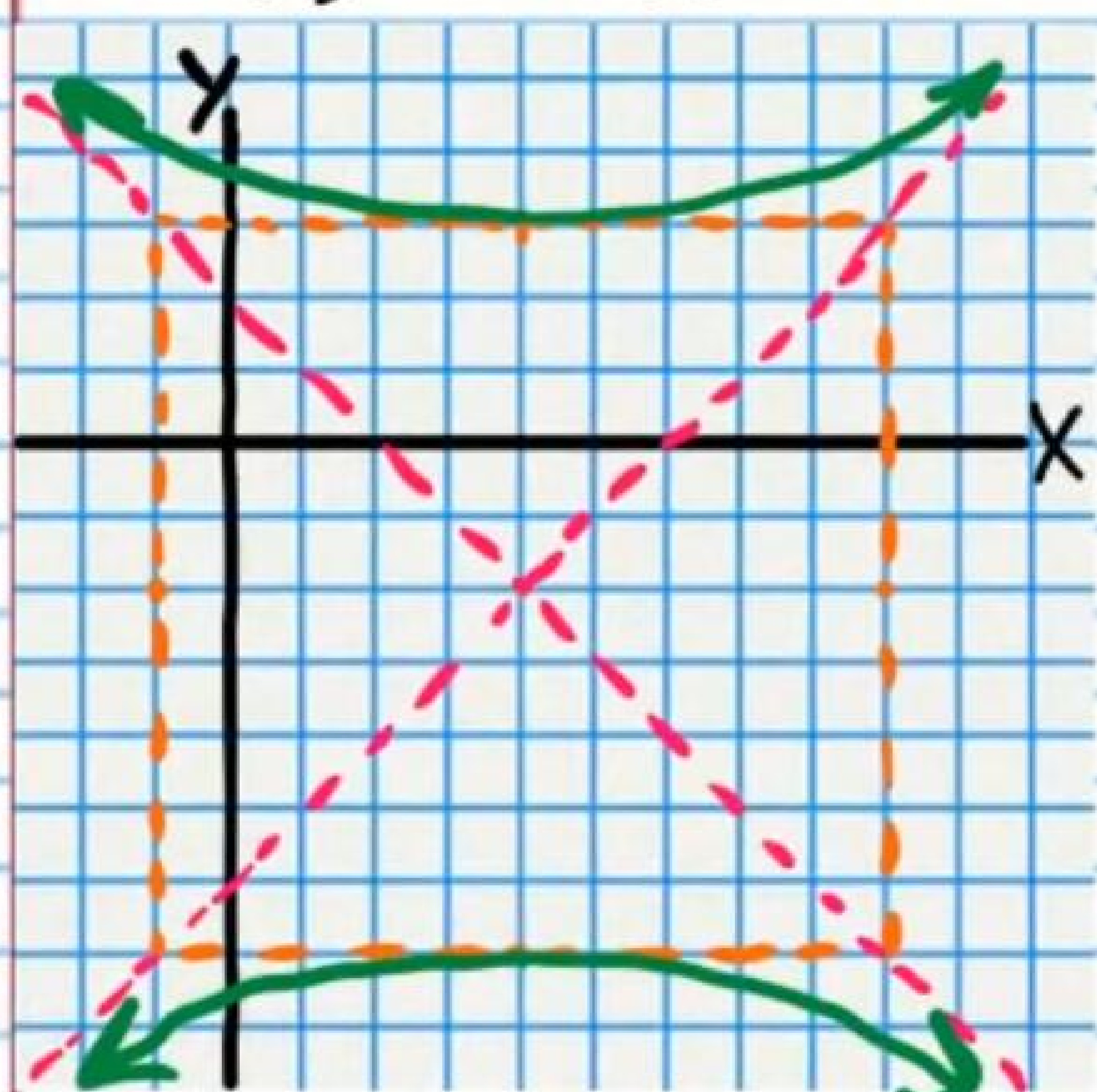
① $(x+5)^2+(y-0)^2=1$



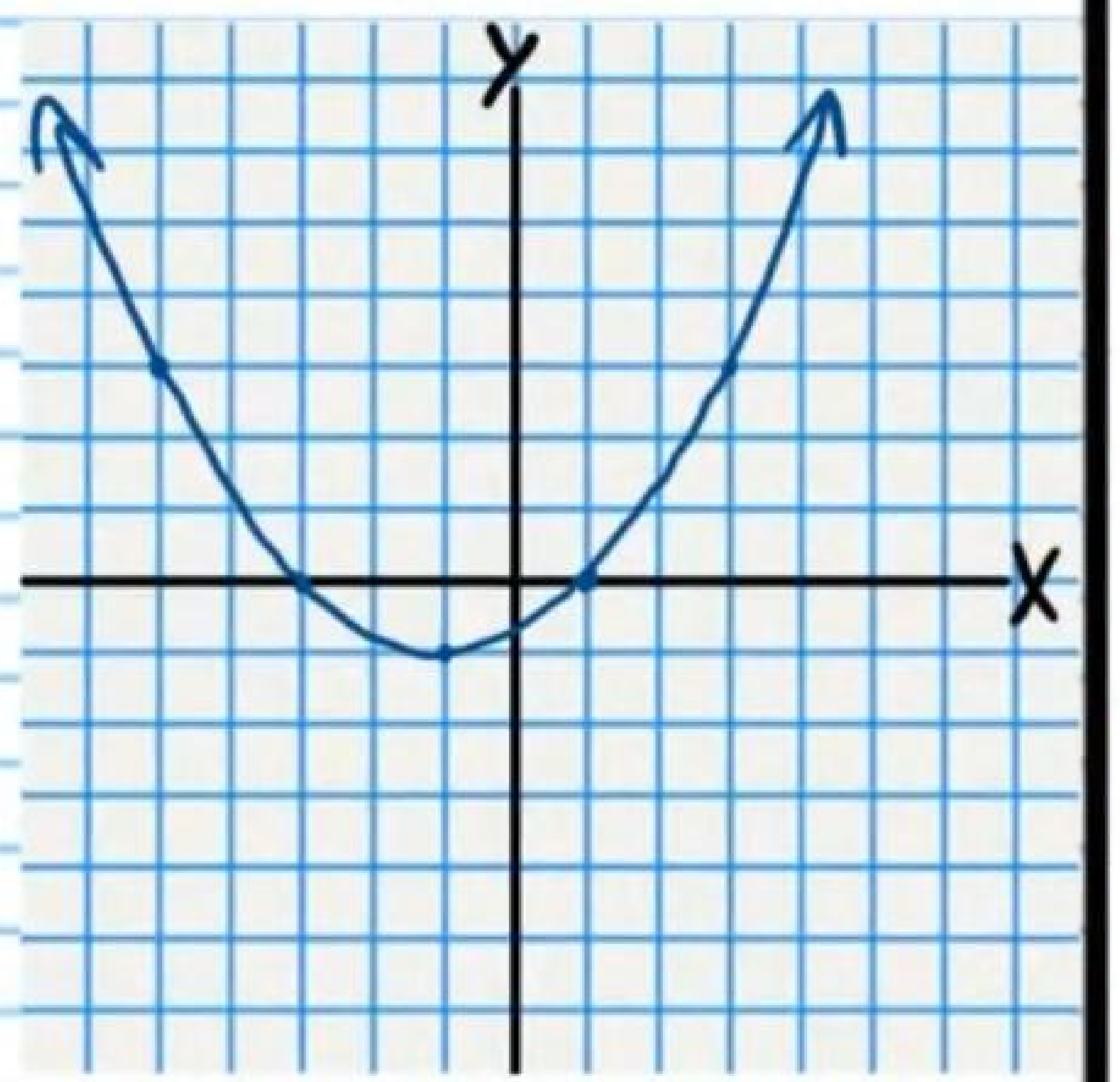
② $\dfrac{(x-3)^2}{9}+\dfrac{(y-1)^2}{16}=1$



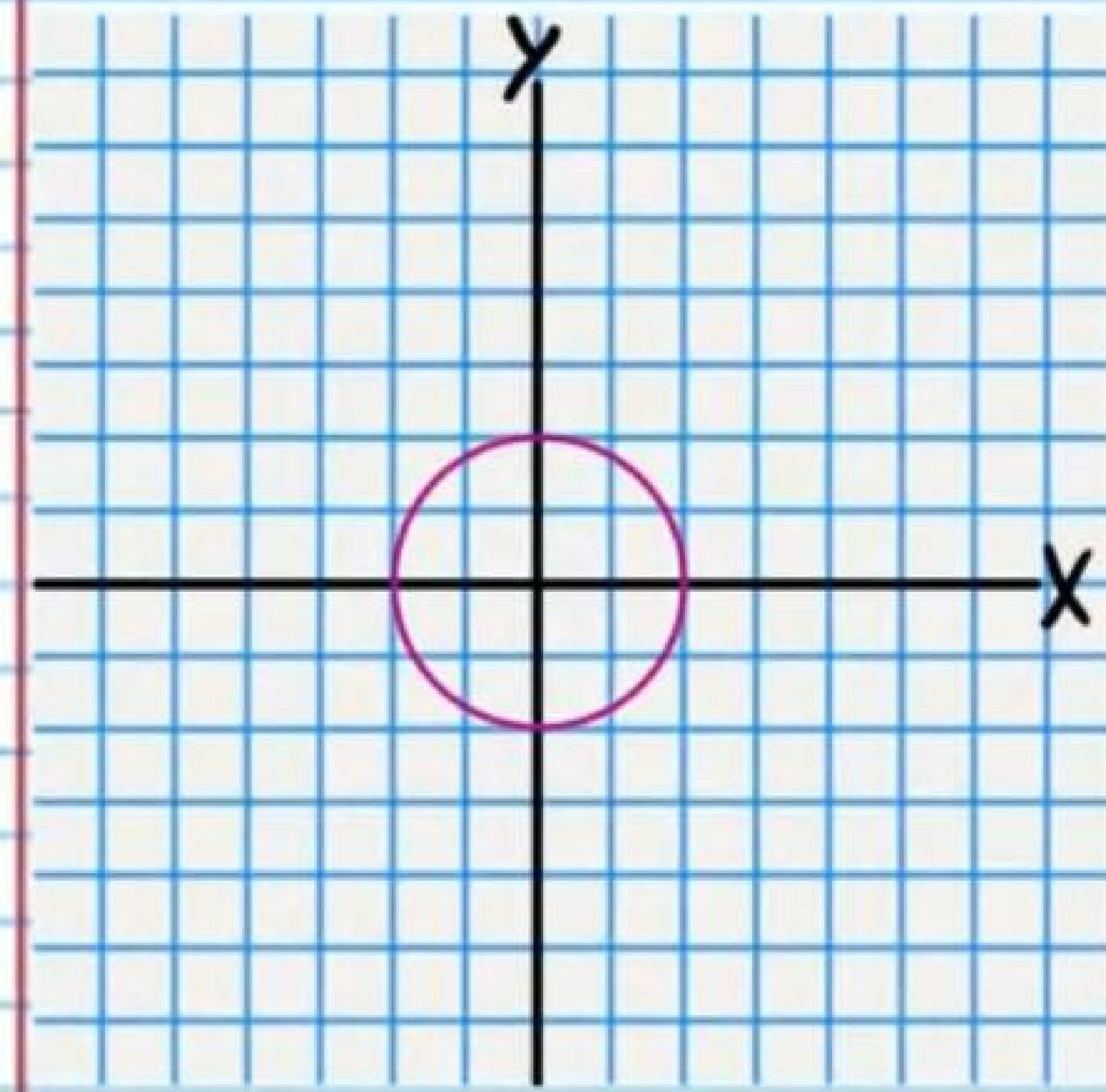
③ $\dfrac{(y+2)^2}{25}-\dfrac{(x-4)^2}{25}=1$



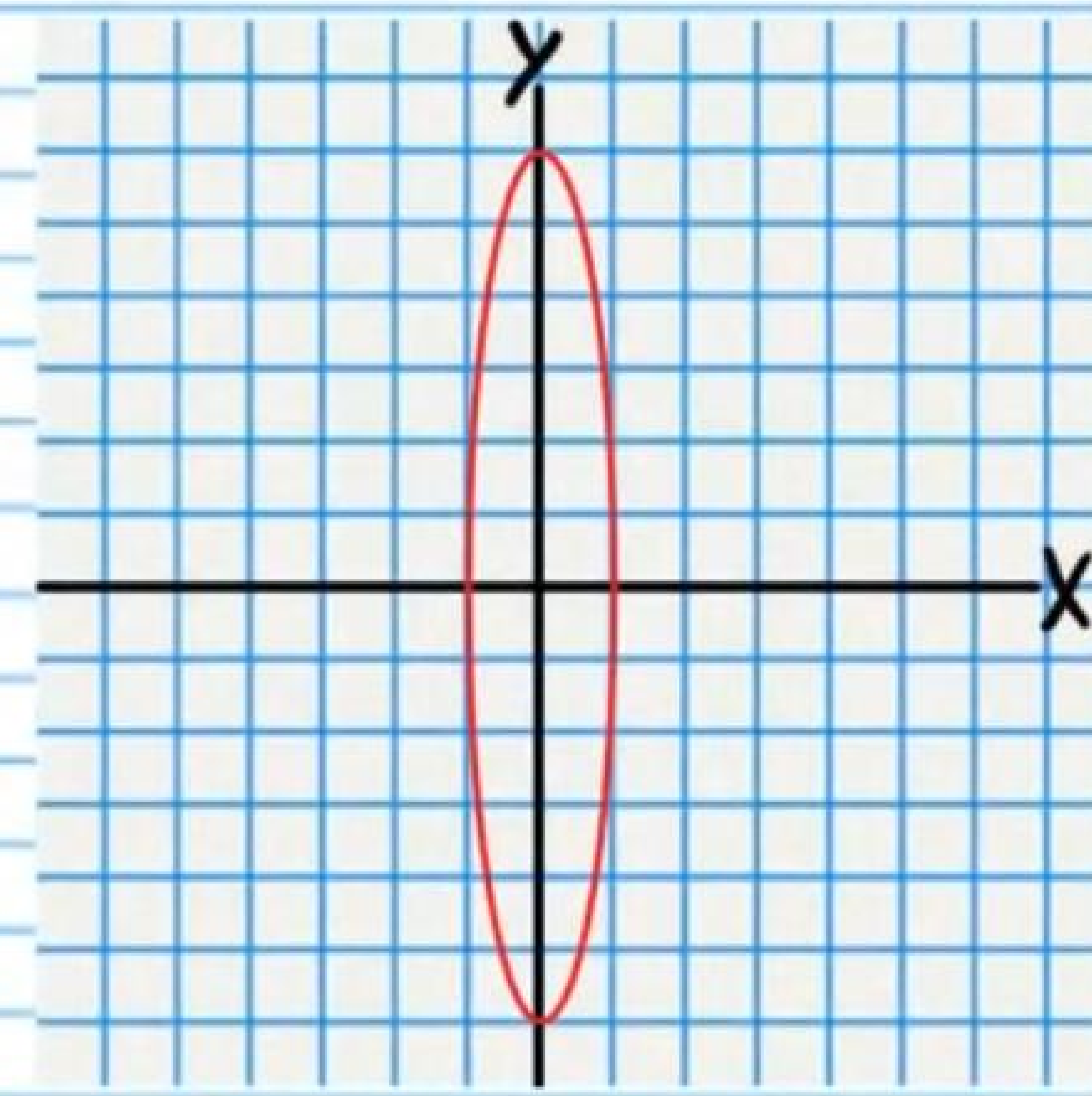
④ $y+1=\frac{1}{4}(x+1)^2$



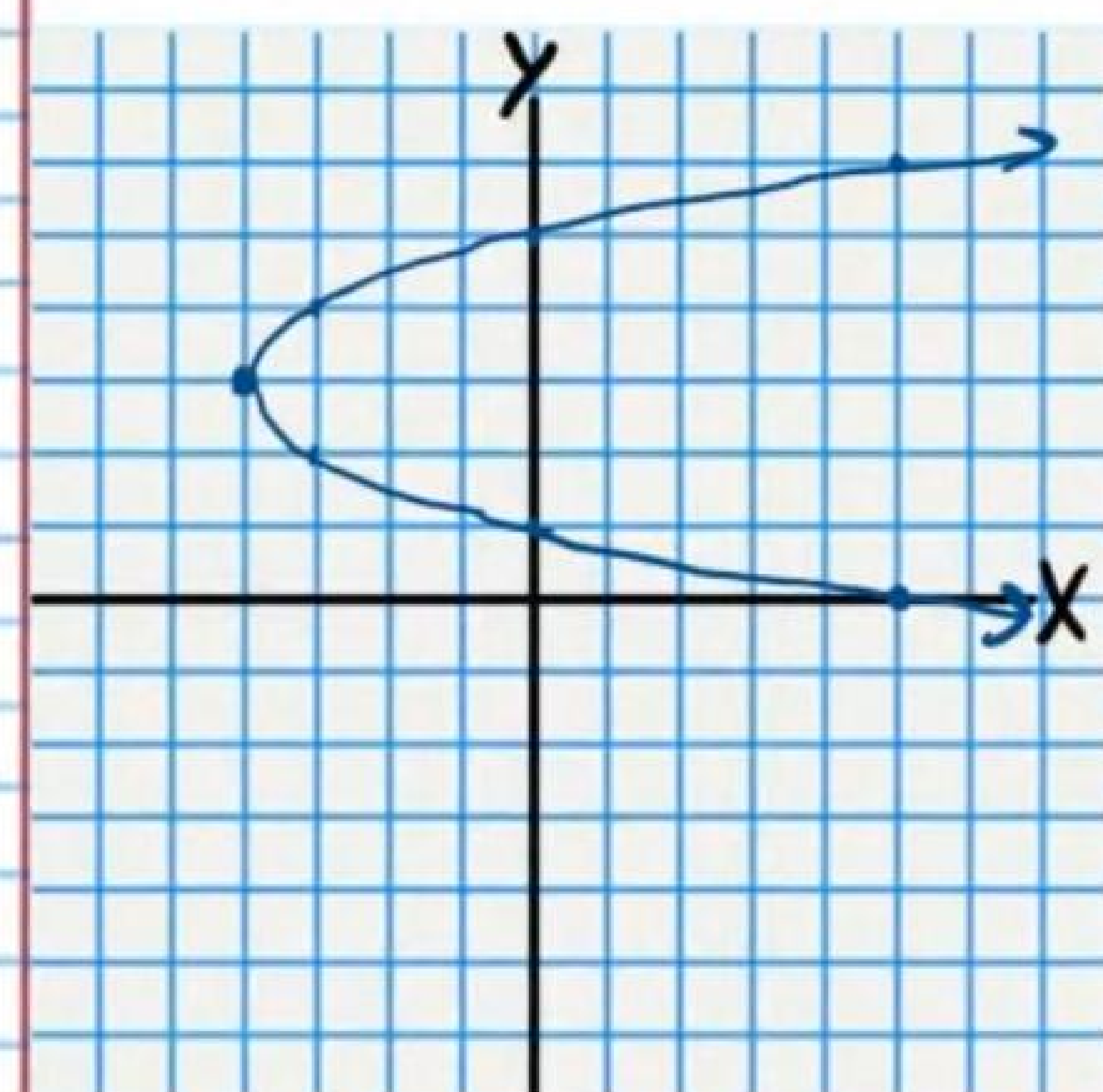
⑤ $(x-0)^2+(y-0)^2=4$



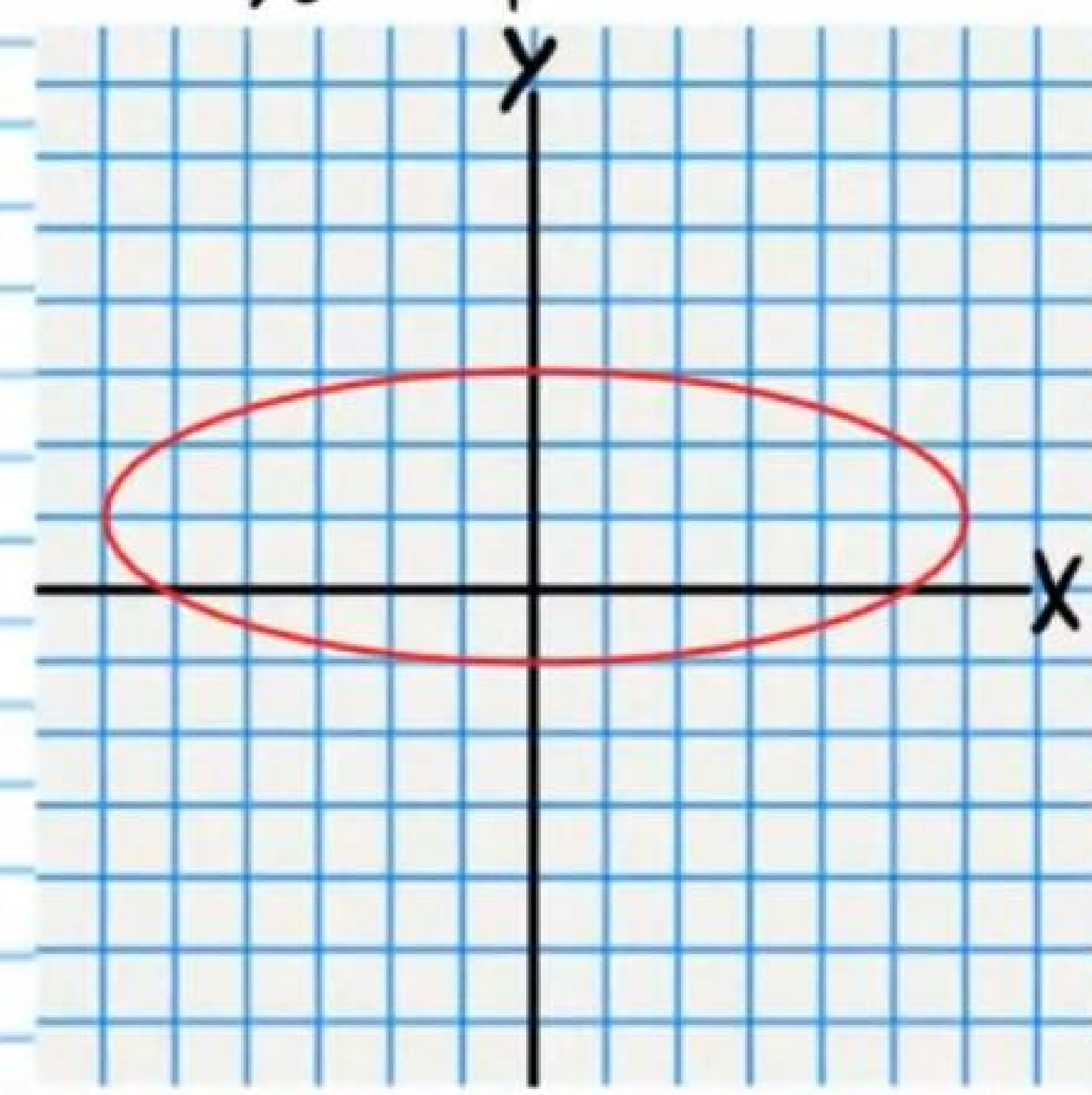
⑥ $(x-0)^2+\dfrac{(y-0)^2}{36}=1$



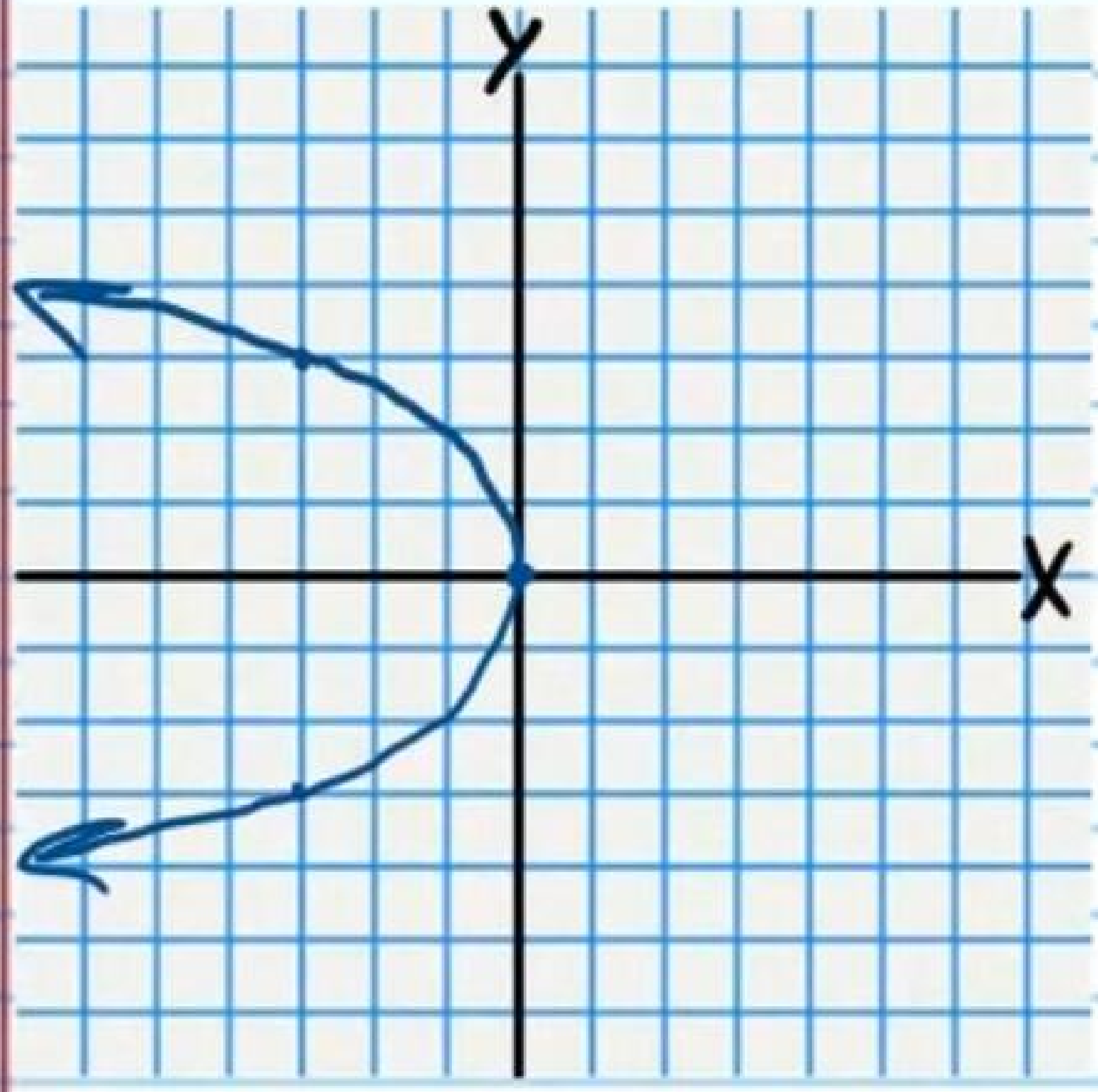
⑦ $x+4=(y-3)^2$



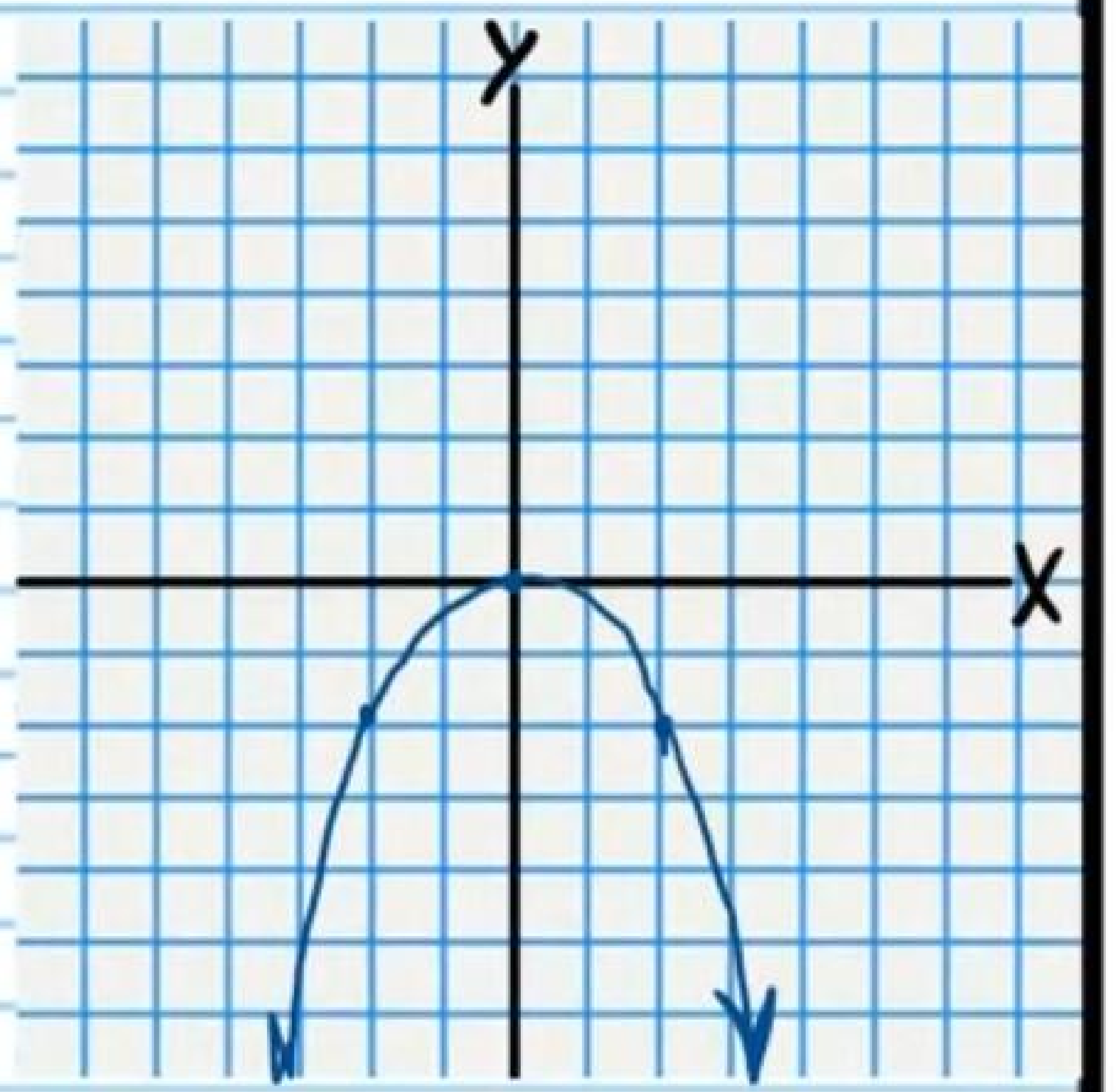
⑧ $\dfrac{(x-0)^2}{36}+\dfrac{(y-1)^2}{4}=1$



⑨ $x-0=-\frac{1}{3}(y-0)^2$



⑩ $y-0=-\frac{1}{2}(x-0)^2$



⑪ $\frac{(x-0)^2}{16}-(y-4)^2=1$